

Minimax Theorem “John von Neuman”

The Minimax Theorem states:

In a two-player, zero-sum game, there exists an optimal strategy for each player such that one player **minimizes the maximum possible loss** while the other **maximizes the minimum guaranteed gain**.

This means:

- **Player A** tries to maximize the minimum payoff they can guarantee.
- **Player B** tries to minimize the maximum payoff Player A can receive.

Eventually, both players reach an **optimal strategy**, where neither can improve their outcome by changing strategies alone.

Note: Both of them wants to stay in the game, survival”

Player 1: "Which strategy gives me the best outcome even in the worst case?"

Player2: Find the **maximum payoff** A can obtain for each of B's strategies.

Choose the **minimum** of those values.

NOTE: Here we an observe both the focus are on **Player A**.

Player A considering about itself and thinking in that perspective.

While Player B also focuses on Player A and tries to minimize the max gain of A.

Mathematical Statement

For a payoff matrix A :

$$\max_x \min_y x^T A y = \min_y \max_x x^T A y$$

Where:

- x = strategy of Player A
- y = strategy of Player B
- A = payoff matrix

The theorem says:

The maximum of the minimum payoffs equals the minimum of the maximum payoffs.

EXAMPLE: SADDLE POINT

	B1	B2
A1	3	2
A2	5	4

Row minima:

- A1 → 2
- A2 → 4

Maximin = 4

Column maxima:

- B1 → 5
- B2 → 4

Minimax = 4

$$\text{Maximin} = \text{Minimax} = 4$$

there is a **saddle point**, and both players have optimal **pure strategies**:

- Player A chooses **A2**
- Player B chooses **B2**

Intuition

Imagine a chess tournament.

You don't know exactly what your opponent will do, so you prepare for the **strongest possible response**. Rather than aiming for the highest possible score in the best case, you aim to secure the **best guaranteed outcome** if your opponent plays optimally.

This cautious reasoning is the essence of the Minimax Theorem.

Assumptions of the Minimax Theorem

The theorem applies under these conditions:

- Two players
- Zero-sum game (one player's gain equals the other's loss)
- Rational players
- Simultaneous decision-making
- Complete knowledge of the game's rules and payoffs

Real-Life Applications

1. Chess
2. Military Strategy
3. Cybersecurity
4. Business Competition
5. Politics and International Relations